

Optic Disc and Exudates Segmentation on Retinal Fundus Images Using Mask R-CNN

I Md. Dendi Maysanjaya
Virtual Vision, Image, and Pattern
Research Group
Universitas Pendidikan Ganesha
Bali, Indonesia
dendi.ms@undiksha.ac.id

Made Windu Antara Kesiman
Virtual Vision, Image, and Pattern
Research Group
Universitas Pendidikan Ganesha
Bali, Indonesia
antara.kesiman@undiksha.ac.id

I G.A.A. Diatri Indradewi
Virtual Vision, Image, and Pattern
Research Group
Universitas Pendidikan Ganesha
Bali, Indonesia
indradewi@undiksha.ac.id

Abstract—Diabetic retinopathy (DR) is an illness caused by diabetes. In 2019, the Centers for Disease Control and Prevention (CDC) approximated 1.4 million new illnesses of diabetes. To identify the DR severity, the ophthalmologist checked the appearance of clinical features, and one of them is exudates. Exudates arise due to the accumulation of lipids and proteins due to a ruptured retinal vein. Both exudates and optic disc (OD) are almost identical, so it is necessary to conduct the region segmentation to remove OD while want to detect exudates. To obtain new insight into computer-aided diagnosis (CAD) on DR, this work aims to segment OD and exudates, by implementing the Mask Region-based Convolutional Neural Network (Mask R-CNN) method and the ResNet50 as the backbone network. For the dataset, we used the IDRiD dataset, which contains 54 train images and 27 test images. In this work, we applied Intersection over Union (IoU) as an evaluation parameter. As the result, our proposed scheme gained the IoU value up to 0.8431 and 0.9933 in segmenting OD and exudates on each. It indicates that the applied scheme can support the establishment of CAD for the segmentation of OD and exudates.

Keywords—segmentation, optic disc, exudates, diabetic retinopathy, retinal fundus image, mask r-cnn

I. INTRODUCTION

Diabetic retinopathy (abbreviated by DR) is an illness affected by diabetes. According to a Centers for Disease Control and Prevention (CDC) report, 1.4 million new illnesses of diabetes were estimated to have been spotted among adults (18 and older) in 2019 [1]. Diabetes is defined by persistently high blood glucose levels [2]. There are two types of diabetes in common. It is called type 1 and type 2 diabetes. For chronic patients with diabetes, it will affect their eyes and can lead to blindness. It is caused by the vein injury of the light-sensitive tissue at the retina.

Diabetic retinopathy is divided into Non-Proliferative Diabetic Retinopathy (NPDR) and Proliferative Diabetic Retinopathy (PDR), while the NPDR is split into three severities, i.e., mild, moderate, and severe NPDR [3], [4]. To identify the DR severity level, the ophthalmologist checked the appearance of clinical features, i.e., exudates, cotton wool spots, hemorrhages, microaneurysms, macular edema, and neovascularization, as depicted in Fig. 1.

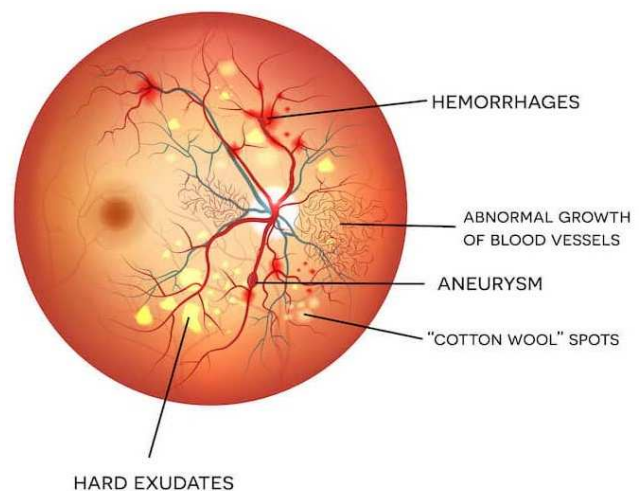


Fig. 1. Symptoms of Diabetic Retinopathy

Exudates are the main clinical symptom used as an early symptom of DR. This condition arises due to the accumulation of lipids and proteins due to ruptured retinal blood vessels. The characteristics of exudates are yellowish, vary in size, and are scattered around the retina. This feature is almost like an optical disc (OD), so OD is necessary to eliminate it first during the detection process of exudates [3]. If it can be done prior, the exudates detection can be easily conducted, because there is no more region in the retinal fundus image that have similar characteristic to exudates.

Hasan, *et al.* [5] proposed the DRNet method for segmenting and locating OD and fovea centers. DRNet was applied to the dataset from IDRiD, RIMONE, DRISHTI-GS, dan DRIVE. Mean Intersection over Union (MIoU) was utilized as a testing parameter. They achieved 0.845, 0.901, 0.933, and 0.920 for IDRiD, RIMONE, DRISHTI-GS, and DRIVE, in sequence.

Aswathy, *et al.* [6] utilized deep learning for DR grading. They offered a hybrid algorithm to recognize DR utilizing semi-supervised semantic segmentation and localizing image-segmentation features. As the result, they gained 97.3% of accuracy. Sun, *et al.* [7] utilized Faster R-CNN to find out the

OD edge on the ORIGA dataset. They achieved 93.1% for the average intersection ratio for the segmentation of OD.

Based on the previous research, the goal in this research area is how to provide the optimum method to help the ophthalmologist identify the DR symptom, and we find out the accuracy of DR symptom detection results can be boosted, especially by using another hybrid method. Therefore, the main purpose of this study focuses to segment OD and exudates by using Mask R-CNN and combining with ResNet50 as the backbone network. We proposed this combination because of the capability of Mask R-CNN to detect the object and make it into the region, so we can detect a different object in a parallel process.

To accomplish the detection work of DR symptoms, we offer a scheme to detect OD and exudates. This paper was organized by structure as follows Section II explained the research approach. The methods are described in Section III

continued by the Result and Discussion in Section IV, then closed by the Conclusion in Section V.

II. APPROACH

Retinopathy Image Analysis (RETINA) is our research project with several milestones, with the first milestone segmenting OD and exudates. To complete our first project milestone, our research flowchart is depicted in Fig. 2. Based on Fig. 2, we split the process for the optic disc segmentation and exudates segmentation from the dataset, so these two processes will run in parallel. Since this work applied Mask R-CNN, we build the model based on the image train on each object (optic disc and exudates). Beside applied Mask R-CNN, we also applied Residual Neural Network (ResNet) 50 as a backbone. In the training process, MaxEpoch is set at 500 epochs.

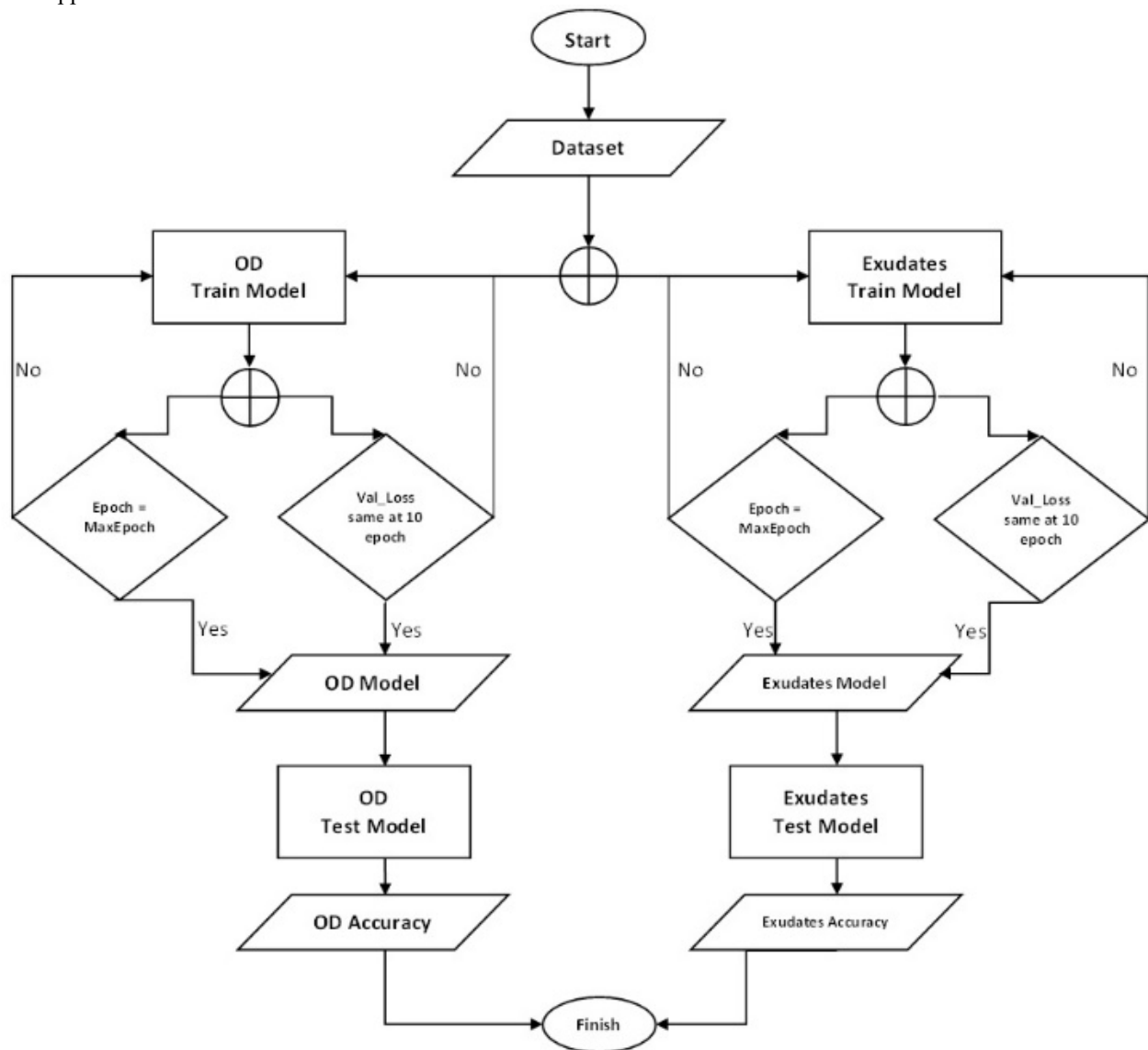


Fig. 2. The Research Flowchart

We set two different conditions to stop the training process; the training process will be stopped if it reached the MaxEpoch or if the validation loss value does not change at 10 epochs in sequence. The output of this step will provide the training model, for each object (OD and exudates), that is ready to use in the testing stage. The training model has then applied to segment the image test and compared the result with the ground truth, then evaluated by Intersection over Union (IoU) parameter.

III. METHODS

This study focuses to segment optic discs and exudates based on diabetic retinopathy retinal fundus images by applying Mask R-CNN. The explanation of data, methods and evaluation parameters will be described in the next part.

A. Dataset

The dataset of this study taken from the Indian Diabetic Retinopathy Image Dataset (IDRiD) [8], [9]. At that source, there are several categories of diabetic retinopathy symptoms, i.e. hard exudates, soft exudates, microaneurysms, hemorrhages, and one of the diabetic retinopathy non-symptoms is the optic disc. In this study, we only used the optic disc and exudates categories.

The dataset comprised 54 images as train sets and 27 as test sets. The dataset is gathered in JPG format, with a size of 4288 x 2848 pixels, and 24 bits of color depth. The sample dataset is shown in Fig. 3.



Fig. 3. Sample of IDRiD Dataset [8]

B. Mask R-CNN

Mask Region-based Convolutional Neural Network, abbreviate as Mask R-CNN, was developed by He *et al.* in 2017, by developing the Faster R-CNN section to predict the segmentation mask for each region of interest (RoI), in parallel with the existing branch for classification and bounding box regression (illustrated in Fig. 4) [10].

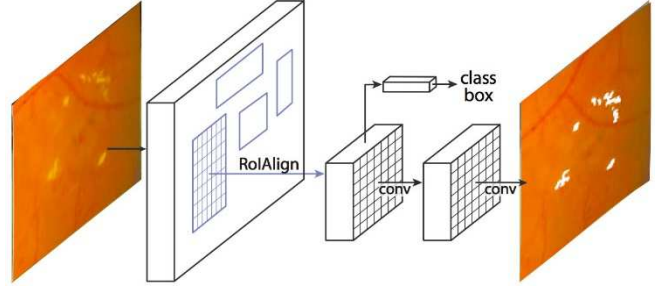


Fig. 4. Illustration of Mask R-CNN [10]

Based on Fig. 4, Mask R-CNN adopted the two stages of Faster R-CNN and added the third branch that outputs the object mask. The first stage is called Region Proposal Network (RPN). RPN is a proposed candidate object bounding box. The second stage, in parallel to divining the class and box offset, outputs a binary mask for each RoI.

In this study, we proposed Mask R-CNN for segmenting the OD and exudates because it can locate each object by the connected area itself, and transform it into several regions based on the object's characteristics.

C. ResNet50

Residual Neural Network (ResNet) 50 is 50 layers of Residual Network proposed by He *et al.* [11], which has 48 convolution layers along with 1 MaxPool and 1 Average Pool layer. It has 3.8×10^9 floating points operations. The convolutional layers size of ResNet50 is 3x3 filters, with the input size fixed as 224x224. Each layer has the same number of filters and output. The filter will be doubled while the output size of the convolved is halved. It ends with an average pooling layer and a 1000-ways fully-connected layer with softmax [12]. Fig. 5 shows the illustration of the ResNet50 architecture.

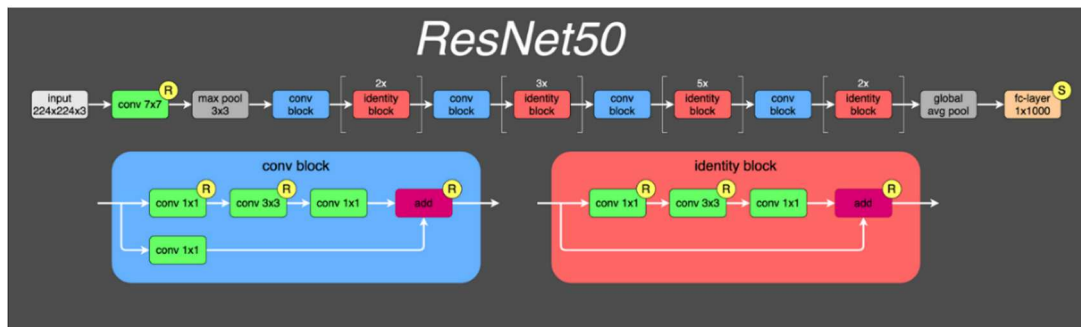


Fig. 5. The ResNet50 Architecture [13]

D. Evaluation Parameter

To evaluate the segmentation result, both optic disc segmentation, and exudates segmentation, the Intersection over Union (IoU) method was utilized. IoU is an evaluation parameter to estimate the segmentation accuracy by calculating the crossing area of the prediction result with the ground truth. This area then divides by the union area both the prediction and the ground truth. The IoU was performed by (1).

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} = \frac{|A \cap B|}{|A \cup B|} \quad (1)$$

where A and B are the prediction and ground truth on each. Besides applied IoU, sensitivity and specificity also were utilized. Sensitivity refers to the classifier's ability to predict the positive class (TP). Meanwhile, specificity is the classifier's ability to recognize the negative class (TN). Both of these parameters are expressed in (2) and (3) respectively.

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (2)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (3)$$

IV. RESULT AND DISCUSSION

In this section, we provide some investigation results of our approach. Table I presented the detail of the Mask R-CNN parameter. Fig. 6 and Fig. 7, show the loss and validation loss for the optic disc train model and the exudates train model respectively.

TABLE I. DETAIL OF THE MASK R-CNN PARAMETER

Parameter	Description
Batch Size	4
Backbone Network	ResNet50
Learning Rate	1x10-4
Max Epoch	500
Step per Epoch	600
Validation Step	100
Input Resolution (pixel)	512x512

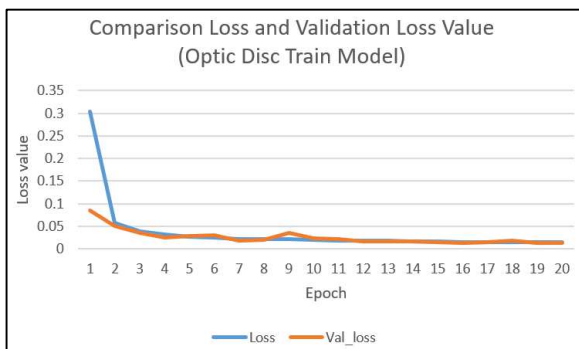


Fig. 6. Comparison of Loss and Validation Loss of OD Train Model

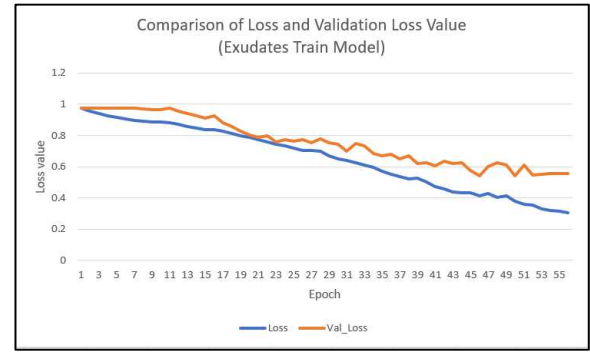


Fig. 7. Comparison of Loss and Validation Loss of Exudates Train Model

Based on Fig. 6, the training process for the OD object stopped at epoch 20 with loss value and validation loss value gained 0.0149 and 0.0128 on each. Meanwhile, in Fig. 7, the training process stopped at epoch 56 with loss value and validation loss value achieved 0.3071 and 0.5540 on each. To validate the performance train model both OD and exudates, we applied it to the test set that contains 27 images, and the results are presented in Table II.

TABLE II. ACCURACY TEST OF THE OPTIC DISC AND EXUDATES TRAIN MODEL

No	Image ID	OD IoU	Exudates IoU
1	IDRiD 55	0.8225	0.9950
2	IDRiD 56	0.8384	0.9953
3	IDRiD 57	0.9204	0.9863
4	IDRiD 58	0.9341	0.9930
5	IDRiD 59	0.8833	0.9987
6	IDRiD 60	0.8754	0.9976
7	IDRiD 61	0.8824	0.9985
8	IDRiD 62	0.7442	0.9957
9	IDRiD 63	0.8398	0.9935
10	IDRiD 64	0.8378	0.9951
...	...	...	...
27	IDRiD 81	0.8797	0.9824
Average		0.8431	0.9933

Based on the test result in Table II, we compared our results to the top five teams (for the optic disc segmentation competition) and the top four teams (for the exudates segmentation competition) based on the IDRiD dataset competition [9] that presented in Table III and Table IV respectively.

TABLE III. COMPARISON OF THE OPTIC DISC ACCURACY

Team	IoU	Method	Input Size (Pixels)	External Dataset
ZJU-BH-SGEX	0.9338	Mask R-CNN	1024x1024	RIGA
VRT	0.9305	U-Net	640x640	DRIVE, DRIONS-DB
IITKgpKLIV	0.8572	SegNet	536x356	Drishti-GS
SDNU	0.7892	Mask R-CNN	1984x1318	-
CBER	0.8912	Handcrafted Features	536x356	-
This work	0.8431	Mask R-CNN	512x512	-

TABLE IV. COMPARISON OF THE EXUDATES ACCURACY

Team	IoU	Method	Input Size (Pixels)	External Dataset
PA Tech	0.8850	DenseNet+U-Net	256x256	-
iFLYTEK	0.8741	Cascaded CNN	320x320	-
SAIHST	0.8582	U-Net	512x512	-
LzyUNCC-1	0.8202	FCN-DLA	1024x1024	E-Ophtha
This work	0.9933	Mask R-CNN	512x512	-

Based on Table III, our work achieved the highest IoU value compared to the SDNU team. We are the same in applying Mask R-CNN but have a difference in the input size. Meanwhile, we could not compare it to the ZJU-BH-SGEX team, since they add an external dataset to their training model. So, our result for OD segmentation still has a contribution, since we resize the dimension of the input size.

Based on Table IV, we gained the highest IoU value than the other teams, even though based on Fig. 7, the validation loss value obtained 0.5540. So, we investigated deeper by analyzing the sensitivity and specificity values. The sensitivity and specificity achievement are provided in Table V.

TABLE V. THE RESULT OF THE SENSITIVITY AND SPECIFICITY TEST

No	Image ID	Sensitivity	Specificity
1	IDRiD_55	0.6254	0.9967
2	IDRiD_56	0.0662	0.9999
3	IDRiD_57	0.7132	0.9921
4	IDRiD_58	0.6719	0.9970
5	IDRiD_59	0.4807	0.9991
6	IDRiD_60	0.5491	0.9984
7	IDRiD_61	0.4000	0.9994
8	IDRiD_62	0.5389	0.9981
9	IDRiD_63	0.2413	0.9988
10	IDRiD_64	0.5611	0.9982
...	...	...	...
27	IDRiD_81	0.3300	0.9989
Average		0.4541	0.9974

Based on Table V, the average of the sensitivity gained is 0.4541, which represents that the training model does not yet have sufficient knowledge to recognize exudates. It means that there are pixels other than exudates that are considered exudates, thus reducing the ability of Mask R-CNN. Based on the results obtained, it is necessary to improve the identifying exudates.

V. CONCLUSION

This work was conducted to detect OD and exudates on the retinal fundus image. The dataset was taken from the IDRiD dataset. The total image of the dataset is 81 images, which divide into 54 images of the training set and 27 images of the testing set. OD and exudates were detected using Mask R-CNN and ResNet50. To get the best training model, two

conditionals to stop the epoch are applied, namely, the training process will be stopped if it reached the MaxEpoch or if the validation loss value does not change at 10 epochs in sequence. By using IoU as an evaluation parameter, our model obtains the IoU value of 0.8431 for OD and 0.9933 for exudates. In the next investigation, we need to explore exudates segmentation, since it has a sensitivity value of about 0.4541 on average. The investigation will be conducted to implement image enhancement prior, so we hope it will bring positive results.

REFERENCES

- [1] Centers for Disease Control and Prevention, "National Diabetes Statistics Report," National Diabetes Statistics Report, Jan. 26, 2022. <https://www.cdc.gov/diabetes/data/statistics-report/index.html> (accessed Aug. 29, 2022).
- [2] Harvard Health Publishing, "Diabetes Mellitus," Drugs.com, Aug. 24, 2021. <https://www.drugs.com/health-guide/diabetes-mellitus.html> (accessed Aug. 15, 2022).
- [3] H. A. Nugroho, K. Z. W. Oktoeberza, T. B. Adji, and M. B. Sasongko, "Segmentation of exudates based on high pass filtering in retinal fundus images," Proceedings - 2015 7th International Conference on Information Technology and Electrical Engineering: Envisioning the Trend of Computer, Information, and Engineering, ICITEE 2015, pp. 436–441, 2015, DOI: 10.1109/ICITEE.2015.7408986.
- [4] D. Lokuarachchi, K. Gunarathna, L. Muthumal, and T. Gamage, "Automated Detection of Exudates in Retinal Images," Proceedings - 2019 IEEE 15th International Colloquium on Signal Processing and its Applications, CSPA 2019, no. March, pp. 43–47, 2019, DOI: 10.1109/CSPA.2019.8696052.
- [5] M. K. Hasan, M. A. Alam, M. T. E. Elahi, S. Roy, and R. Marti, "DRNet: Segmentation and localization of optic disc and Fovea from diabetic retinopathy image," Artif Intell Med, vol. 111, Jan. 2021, DOI: 10.1016/j.artmed.2020.102001.
- [6] Aswathy K.P and D. V. Kumar, "Diabetic Retinopathy Detection Using Semantic Segmentation And Optic Disc Localization," Turkish Journal of Computer and Mathematics Education, vol. 12, no. 9, pp. 305–311, 2021.
- [7] X. Sun, Y. Xu, W. Zhao, T. You, and J. Liu, "Optic Disc Segmentation from Retinal Fundus Images via Deep Object Detection Networks," in 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Oct. 2018, pp. 5954–5957.
- [8] P. Porwal et al., "Indian diabetic retinopathy image dataset (IDRiD): A database for diabetic retinopathy screening research," Data (Basel), vol. 3, no. 3, pp. 1–8, 2018, DOI: 10.3390/data3030025.
- [9] P. Porwal et al., "IDRiD: Diabetic Retinopathy – Segmentation and Grading Challenge," Med Image Anal, vol. 59, 2020, DOI: 10.1016/j.media.2019.101561.
- [10] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in 2017 IEEE International Conference on Computer Vision (ICCV), Dec. 2017, pp. 1–12. [Online]. Available: <http://arxiv.org/abs/1703.06870>
- [11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Dec. 2016, vol. 2016-December, pp. 770–778. DOI: 10.1109/CVPR.2016.90.
- [12] R. I. Bendjillali, M. Beladgham, K. Merit, and A. Taleb-Ahmed, "Illumination-robust face recognition based on deep convolutional neural networks architectures," Indonesian Journal of Electrical Engineering and Computer Science, vol. 18, no. 2, pp. 1015–1027, 2020, DOI: 10.11591/ijeecs.v18.i2.pp1015-1027.
- [13] R. Karim, "Illustrated: 10 CNN Architectures," Towards Data Science, Jul. 30, 2019. <https://towardsdatascience.com/illustrated-10-cnn-architectures-95d78ace614d> (accessed Sep. 27, 2022).